

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	40% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	2024-AI&ML	Date:	17.08.2026

Code of Conduct

I D.Sravanthi(192425315) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: D.Sravanthi

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.

A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

1. Analyze the semantic representations and identify the action-object relationships.

Query	Semantic Representation	Action	Object
Activate international roaming	ACTIVATE(Roaming, Customer)	Activate	Roaming
Deactivate caller tune	DEACTIVATE(CallerTune, Customer)	Deactivate	Caller Tune
Check data balance	QUERY(DataBalance, Customer)	Query	Data Balance
Enable 5G service	ACTIVATE(5GService, Customer)	Activate	5G Service

2. Determine which query contains semantic interpretation errors.

- **Q1:** Correct – Activate Roaming
- **Q2: Incorrect** – Actual: Deactivate Caller Tune; Predicted: Activate Caller Tune
- **Q3:** Correct – Query Data Balance
- **Q4:** Correct – Activate 5G Service

Therefore, **Q2 contains the semantic interpretation error**. The system misunderstood *deactivate* as *activate*.

3. Evaluate how meaning representation affects chatbot decision accuracy.

Meaning representation directly affects the chatbot's final decision. If the action or object is incorrectly identified, the chatbot may perform the wrong operation. For example, interpreting **DEACTIVATE(CallerTune, Customer)** as activation could result in activating a service that the customer wanted to stop.

4. Recommend improvements to enhance semantic understanding in telecom customer support systems.

Improvements:

- Use context-aware semantic parsing.
- Train the model with more telecom-specific conversations.
- Distinguish clearly between opposite actions such as **activate/deactivate**.
- Use intent classification with entity recognition.
- Add confidence scores and clarification questions for uncertain queries.
- Continuously learn from customer feedback and failed conversations.

2. First-Order Predicate Calculus for Smart Manufacturing.

An Industry 4.0 manufacturing plant monitors machines using semantic rules.

Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

1. Represent the production data using First-Order Predicate Calculus.

Represent production data using First-Order Predicate Calculus

The production data can be represented as:

- **Active(M1)**
- **Active(M2)**
- **Maintenance(M3)**
- **Active(M4)**

Given rules:

- $\forall x \text{ Active}(x) \rightarrow \text{Producing}(x)$
- $\forall x \forall y \text{ Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\forall x \text{ Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Therefore:

- $\text{Producing}(M1)$
- $\text{Producing}(M2)$
- $\neg \text{Producing}(M3)$
- $\text{Producing}(M4)$

2. Analyze which products are currently available.

The data does not specify which machines produce which products. Therefore, **specific products cannot be determined directly.**

However, using the rule:

$\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$

any product produced by **M1, M2, or M4** is considered available.

Products produced by **M3 are not considered available**, because M3 is under maintenance.

3. Infer whether Gear production is affected by maintenance activities.

If **M3 produces Gear**, then:

Maintenance(M3) $\rightarrow$ $\neg$ Producing(M3)

Therefore, Gear production is affected and temporarily stopped.

So, if M3 is the machine responsible for Gear production, Gear production is affected by maintenance.

4. Evaluate the effectiveness of predicate logic in industrial decision-support systems.

Effectiveness of predicate logic

Predicate logic is useful in industrial decision-support systems because it:

- Represents machine states precisely.
- Supports automatic reasoning and inference.
- Detects production constraints.
- Helps identify unavailable machines/products.
- Provides explainable decision rules.
- Can be integrated with expert systems and AI-based monitoring.

A limitation is that real manufacturing environments are dynamic, so predicate rules need to be continuously updated with real-time sensor data.

3. Word Sense Disambiguation in E-Commerce Search Engines.

An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.

Query	Correct Sense	Evidence
Apple accessories	Technology Brand	iPhone Charger
Mouse wireless	Computer Device	Bluetooth Mouse
Java tutorial	Programming Language	Coding Lessons
Python course	Programming Language	Software Development Training

2. Identify the semantic cues used for disambiguation.

Semantic cues used for disambiguation

Important cues include:

- **Apple** → **accessories** → **iPhone Charger** → technology brand.
- **Mouse** → **wireless** → **Bluetooth Mouse** → computer device.
- **Java** → **tutorial** → **Coding Lessons** → programming language.
- **Python** → **course** → **Software Development Training** → programming language.

The system uses **surrounding words, query context, clicked results, and user behavior** to determine meaning.

3. Evaluate the impact of incorrect sense selection on search engine performance.

Impact of incorrect sense selection

Incorrect WSD can cause:

- Irrelevant search results.

- Lower click-through rate.
- Poor product recommendations.
- Reduced customer satisfaction.
- Increased search abandonment.
- Loss of sales and revenue.

For example, interpreting **"Python course"** as a snake would produce completely irrelevant products and information.

4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

Industrial-scale WSD strategy

An e-commerce search engine can use a combination of:

1. **Transformer-based models** such as BERT-style contextual language models.
2. Query context and surrounding words.
3. User click and purchase history.
4. Product titles and descriptions.
5. Knowledge graphs connecting products and concepts.
6. Search-session information.
7. Continuous feedback from clicks, purchases, and user corrections.

A hybrid **contextual NLP + user behavior + knowledge graph** approach can provide highly accurate large-scale WSD.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.

Syntactic structure identifies the relationship between words in a sentence.

For example:

"Doctor prescribed medicine to patient."

- Doctor → Subject → **Agent**
- Prescribed → Verb → Action
- Medicine → Object → **Theme**
- Patient → Indirect object → **Recipient**

Similarly:

"Patient reported severe headache."

- Patient → Subject → **Experiencer**
- Headache → Object/semantic content → **Symptom**

Thus, grammatical relationships such as **subject, object, and indirect object** help the NLP system assign semantic roles.

2. Determine whether the assigned semantic roles are appropriate.

Some roles are appropriate, but there are important corrections:

Entity	Given Role	Correct/Preferred Role
Doctor	Agent	Agent ✓
Medicine	Instrument	Theme

Entity	Given Role	Correct/Preferred Role
Patient	Recipient	Recipient ✓ in the prescription sentence
Headache	Symptom	Symptom ✓

- The major error is **Medicine** → **Instrument**. In *"Doctor prescribed medicine to patient,"* medicine is the **Theme**, meaning the entity being prescribed. It is not an instrument used by the doctor to perform the action.

Potential errors from incorrect parsing

Incorrect syntactic parsing may lead to:

- Wrong identification of doctor and patient roles.
- Incorrect medication relationships.
- Misinterpretation of symptoms.
- Incorrect extraction of treatment information.
- Wrong medical knowledge representation.
- Errors in clinical decision-support systems.

For example, confusing **"Doctor prescribed medicine to patient"** could cause the system to associate the wrong person with the treatment.

- Also, semantic roles can change depending on the sentence. In *"Medicine reduced blood pressure,"* medicine acts as the **Cause/Agent** of the reduction, while blood pressure is the **Theme** or affected entity.

Methods to improve syntax-driven semantic analysis

Healthcare NLP systems should use:

- Medical-domain-trained NLP models.
- Dependency parsing and semantic role labeling.
- Clinical language datasets for training.
- Medical dictionaries and ontologies such as UMLS/SNOMED CT.
- Context-aware transformer models.
- Negation and uncertainty detection.
- Human expert validation for critical cases.
- Continuous evaluation using real clinical reports.

Conclusion: Syntax-driven semantic analysis is highly useful in healthcare because correct grammatical parsing helps identify **who performed an action, what was affected, and what medical entities are involved**. However, it should be combined with domain knowledge and contextual NLP to reduce errors.

- Evaluate potential errors that could arise from incorrect parsing.

Incorrect syntactic parsing can cause the NLP system to misunderstand relationships between medical entities. This may result in:

- **Wrong semantic roles:** Doctor, patient, medicine, or symptom may be assigned incorrect roles.

- **Incorrect treatment information:** The system may misunderstand who prescribed or received a medicine.
- **Wrong symptom identification:** Symptoms may be associated with the wrong patient or condition.
- **Misinterpretation of medical relationships:** For example, confusing the medicine with the instrument used for treatment.
- **Clinical decision errors:** Incorrectly extracted information can affect diagnosis, treatment recommendations, and patient records.
- **Loss of important context:** Negation or uncertainty may be misunderstood, such as “*Patient has no headache.*”

Example: If “*Doctor prescribed medicine to patient*” is parsed incorrectly, the system might identify the medicine as the Agent instead of the Doctor, leading to an incorrect medical interpretation.

6. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

The following methods can improve accuracy:

1. **Use medical-domain NLP models** trained on clinical documents.
2. **Apply dependency parsing** to correctly identify relationships between words.
3. **Use Semantic Role Labeling (SRL)** to identify Agent, Patient, Theme, Symptom, etc.
4. **Use medical ontologies** such as UMLS and SNOMED CT to understand medical concepts.
5. **Handle negation and uncertainty**, such as “no fever” or “possible infection.”
6. **Use transformer-based models** such as BERT-based clinical NLP models for better contextual understanding.
7. **Train with large annotated medical datasets** containing real clinical sentences.
8. **Human expert validation** should be used for critical healthcare decisions.

Overall: Combining syntactic parsing, semantic role labeling, medical knowledge bases, and context-aware AI models can significantly improve the accuracy and reliability of healthcare NLP systems.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____